

# Practical Task 1

1. State whether the mean, median, or mode would be useful in the following scenarios:

- You are doing population statistics. You are asked to give an estimate of the typical income of a single person in the country. There is one snag: wealth distribution is out of whack, and 10% of the population holds 70% of the nation's wealth.

Median, it represents the middle which we could do something meaningful with

- You are running a restaurant, and you are reviewing your menu. You have a list of all orders over the last six months. You are trying to find out which item you should keep based on what customers seem to like the most.

Mode, because it tells us what gets ordered the most

- You have been buying electricity once a month for the first six months of the year. You are trying to budget your electricity for the rest of the year and therefore need to estimate how much you will spend for the remainder of the year.

Mean, because it will help us budget for outliers

- You work in healthcare insurance. You are asked to provide an estimate of the typical amount of money spent on healthcare. This is taking into account the fact that there are a few people who spend a large amount of money on medical healthcare due to major issues.

Median, it could represent the middle which we could do something meaningful with

1. State whether you would use variance or standard deviation to inform the following decisions:

- You are choosing a new Internet provider. You find two providers with the same mean speed, but you want to have a more stable connection. You get a list of all reported speeds over the last month and are trying to find the provider that doesn't move too much from the mean value.

Variance, because it shows us the 'spread'

- You are going on holiday to Mauritius. You need to find a shuttle from the airport to your hotel, but you are worried about being overcharged or undercharged (being undercharged might mean that you get unreliable transport). You get a list of all available shuttle service prices and need to find out which services, if any, are overcharging or undercharging.

Standard deviation, because it shows us the standard boundaries in which the prices should be